

Hiring Process Analytics

Project Description

The project was designed to conduct comprehensive data analytics on the hiring process within a company, with the primary objective of gaining insights into various aspects such as gender distribution, salary analysis, departmental composition, and position tiers. By analyzing these factors, the project aimed to provide valuable insights into hiring patterns and organizational dynamics.

Approach

To accomplish the project objectives, a systematic approach was followed. A dataset containing relevant information on hires, including gender, salary, department, and position, was obtained. Microsoft Excel 2022 was selected as the primary tool for data analysis due to its versatility and robust capabilities in handling tabular data. Specific techniques such as pivot tables, charts, and formulas were employed to analyze the dataset and extract meaningful insights.

Tech-Stack Used

Software: Microsoft Excel 2022

Purpose:

Excel was chosen for its extensive data analysis functionalities, including pivot tables, charts, and statistical functions. It facilitated in-depth exploration and visualization of the hiring data, enabling a thorough understanding of the underlying trends and patterns.

Insights

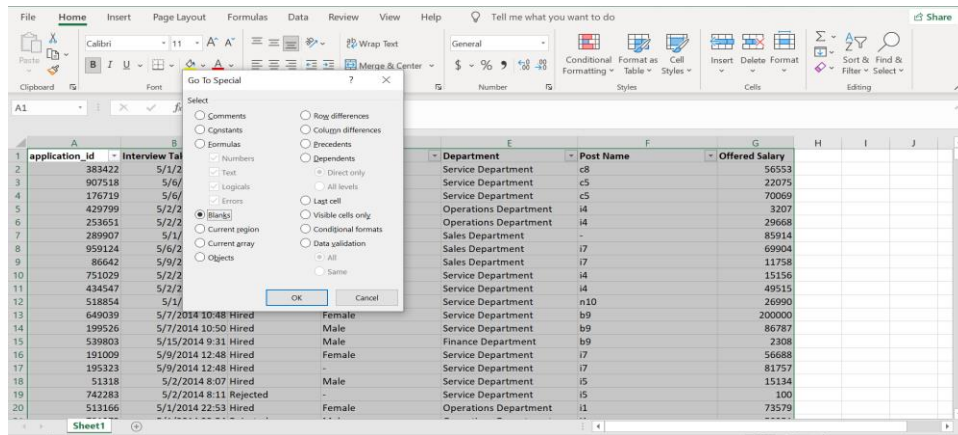
Through the data analytics process, several key insights were uncovered:

- Analysis of gender distribution provided insight into hiring diversity.
- Salary analysis revealed patterns in employee compensation.
- Examination of departmental composition highlighted workforce distribution across departments.
- Analysis of position tiers uncovered the structure of roles within the organization.

Meaningful trends and patterns were observed in the data, shedding light on the company's hiring process and organizational structure.

THERE ARE SOME STEPS TO MODIFY THE SHEET:

- First we have to check the blank in the sheet.



- There is a one blank in the sheet .
- The status is Rejected.
- So, I am deleting the row.

	A	B	C	D	E	F	G	H	I	J
72	188820	5/7/2014 17:58	Rejected	Male	Sales Department	i4	94869			
73	193244	5/7/2014 10:53	Rejected	Male	Operations Department	i6	77157			
74	781620	5/7/2014 15:34	Rejected	Male	Production Department	i4	62894			
75	379557	5/7/2014 15:34	Hired	Female	Production Department	i4	61532			
76	674051	5/7/2014 15:35	Rejected	Female	Production Department	i4	81261			
77	928902	5/7/2014 16:26	Rejected	Female	Finance Department	b9	59644			
78	22155	5/7/2014 16:26	Rejected	Female	Finance Department	b9	46852			
79	572422	5/7/2014 8:07	Rejected	Male	Sales Department	i7	61488			
80	114584	5/7/2014 8:08	Rejected	Male	Sales Department	i7				
81	794226	5/7/2014 16:01	Rejected	Male	Service Department	i7	16090			
82	496573	5/7/2014 16:01	Rejected	Male	Service Department	i7	83364			
83	437695	5/13/2014 15:22	Rejected	Male	Service Department	b9	77517			
84	983577	5/8/2014 7:13	Rejected	Male	Service Department	b9	84746			
85	720850	5/8/2014 22:10	Rejected	Male	Sales Department	i6	80600			
86	453498	5/8/2014 3:12	Hired	Male	Sales Department	i4	62937			
87	550914	5/8/2014 3:15	Rejected	Female	Sales Department	i4	18921			
88	37906	5/8/2014 3:16	Hired	Female	Sales Department	i4	95603			
89	692080	5/8/2014 3:16	Hired	Male	Sales Department	i4	79230			
90	735087	5/8/2014 3:16	Rejected	Male	Sales Department	i4	56650			
91	855558	5/8/2014 11:52	Rejected	Female	Production Department	c5	75158			

	A	B	C	D	E	F	G	H	I	J
1	application_id	Interview Taken on	Status	event_name	Department	Post Name	Offered Salary			
80	114584	5/7/2014 8:08	Rejected	Male	Sales Department	i7				
7170										
7171										
7172										
7173										
7174										
7175										
7176										
7177										

- After we have checked the post name there is no value.
- So, we will modify the post name and department and add the post name.

	A	B	C	D	E	F	G	H	I	J
1	application_id	Interview Taken on	Status	event_name	Department	Post Name	Offered Salary			
7	289907	5/1/2014 7:44	Hired	Male	Sales Department	-	85914			
7170										
7171										
7172										
7173										
7174										
7175										

- There is an empty blank in post name.

	A	B	C	D	E	F	G	H	I	J
1	application_id	Interview Taken on	Status	event_name	Department	Post Name	Offered Salary			
7	289907	5/1/2014 7:44	Hired	Male	Sales Department	-	85914			
1323	882740	8/7/2014 13:42	Rejected	Male	Sales Department	c9	85837			
1457	230796	8/25/2014 9:32	Hired	Female	Sales Department	c9	85569			
4224	226229	7/10/2014 19:03	Hired	Male	Sales Department	c5	85610			
5043	254287	8/13/2014 17:44	Rejected	Male	Sales Department	c5	86346			
7079	270561	7/20/2014 7:14	Hired	Male	Sales Department	i7	86238			
7170										
7171										
7172										

	A	B	C	D	E	F	G	H	I	J
1	application_id	Interview Taken on	Status	event_name	Department	Post Name	Offered Salary			
7133	935497	8/27/2014 17:36	Rejected	Male	Service Department	c5	67196			
7134	106032	8/22/2014 15:50	Hired	Male	Service Department	i7	16756			
7135	166185	8/27/2014 5:01	Hired	Female	Service Department	i7	30952			
7136	736189	8/28/2014 17:29	Hired	Male	Service Department	c9	64150			
7137	614594	8/28/2014 17:30	Hired	Male	Service Department	c9	40152			
7138	493131	8/28/2014 17:32	Hired	Male	Service Department	c9	49282			
7139	214261	8/31/2014 1:36	Hired	Female	Service Department	c5	57742			
7140	932441	8/31/2014 1:37	Hired	Male	Service Department	c5	69932			
7141	39010	8/31/2014 1:38	Rejected	Male	Service Department	c5	14489			
7142	686055	8/26/2014 12:14	Hired	Male	Operations Department	c5	54201			
7143										
7144										
7145										
7146										
7147										
7148										
7149										
7150										
7151										
7152										

- According to the below image c9 is more so, I am adding c9 to the empty blank.

- In this event name the is a blank.
- So, we have to modify the empty blank.

A	B	C	D	E	F	G	H
application_id	Interview Taken on	Status	event_name	Department	Post Name	Offered Salary	
195323	5/9/2014 12:48	Hired	-	Service Department	i7	81757	
742283	5/2/2014 8:11	Rejected	-	Service Department	i5	100	
227046	8/27/2014 18:08	Hired	-	Operations Department	b9	76730	
711350	7/16/2014 13:33	Rejected	-	Operations Department	c-10	25785	
835053	5/16/2014 18:34	Hired	-	Operations Department	c5	25583	
444043	7/11/2014 14:52	Hired	-	Sales Department	c5	80262	
352309	8/20/2014 10:38	Hired	-	Service Department	i5	4308	
204014	8/9/2014 16:09	Rejected	-	Purchase Department	c5	96396	
901867	8/18/2014 9:36	Rejected	-	Service Department	c5	22393	
937905	8/8/2014 19:29	Hired	-	Marketing Department	c9	94032	
564743	8/28/2014 10:25	Rejected	-	Production Department	c9	4076	
245473	5/14/2014 18:48	Hired	-	Service Department	c5	66948	
411295	6/22/2014 14:38	Hired	-	Operations Department	i1	98070	
487617	5/30/2014 16:29	Hired	-	Service Department	c8	12470	
827628	8/30/2014 15:51	Hired	-	Service Department	i1	3134	

- I am adding “Don’t want to say”.
- In the empty blank.

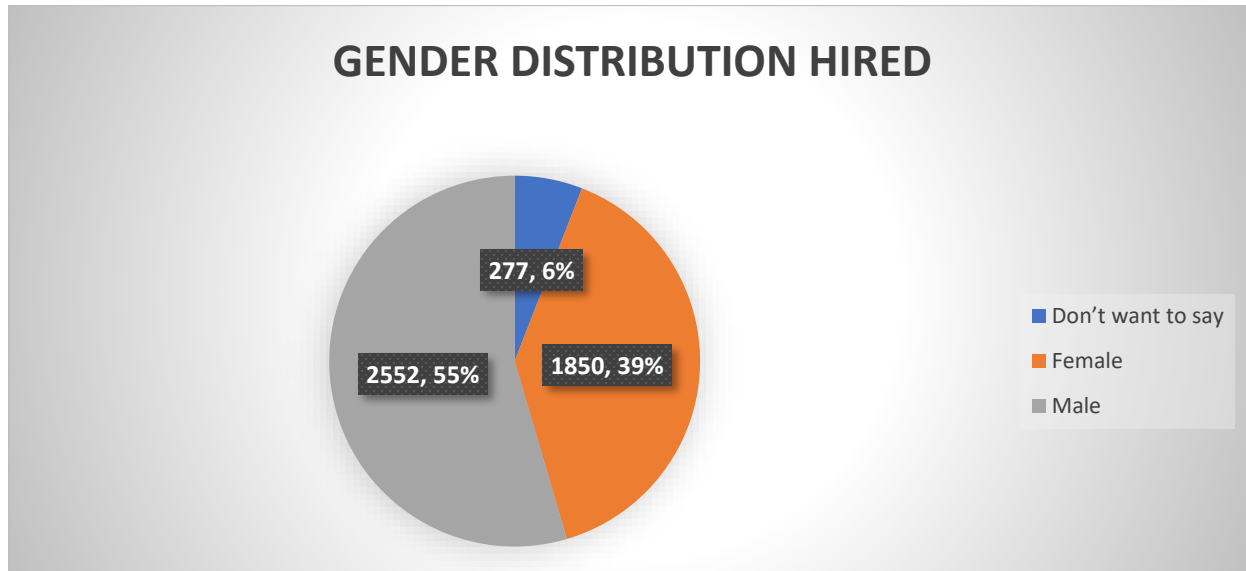
A	B	C	D	E	F	G	H	I	J
application_id	Interview Taken on	Status	event_name	Department	Post Name	Offered Salary			
195323	5/9/2014 12:48	Hired	Don't want to say	Service Department	i7	81757			
742283	5/2/2014 8:11	Rejected	Don't want to say	Service Department	i5	100			
227046	8/27/2014 18:08	Hired	Don't want to say	Operations Department	b9	76730			
711350	7/16/2014 13:33	Rejected	Don't want to say	Operations Department	c10	25785			
835053	5/16/2014 18:34	Hired	Don't want to say	Operations Department	c5	25583			
444043	7/11/2014 14:52	Hired	Don't want to say	Sales Department	c5	80262			
352309	8/20/2014 10:38	Hired	Don't want to say	Service Department	i5	4308			
204014	8/9/2014 16:09	Rejected	Don't want to say	Purchase Department	c5	96396			
901867	8/18/2014 9:36	Rejected	Don't want to say	Service Department	c5	22393			
937905	8/8/2014 19:29	Hired	Don't want to say	Marketing Department	c9	94032			
564743	8/28/2014 10:25	Rejected	Don't want to say	Production Department	c9	4076			
245473	5/14/2014 18:48	Hired	Don't want to say	Service Department	c5	66948			
411295	6/22/2014 14:38	Hired	Don't want to say	Operations Department	i1	98070			
487617	5/30/2014 16:29	Hired	Don't want to say	Service Department	c8	12470			
827628	8/30/2014 15:51	Hired	Don't want to say	Service Department	i1	3134			

A)Hiring Analysis:

The hiring process involves bringing new individuals into the organization for various roles.

Your Task: Determine the gender distribution of hires. How many males and females have been hired by the company

Row Labels	Count of Status
Don't want to say	277
Female	1850
Male	2552
Grand Total	4679

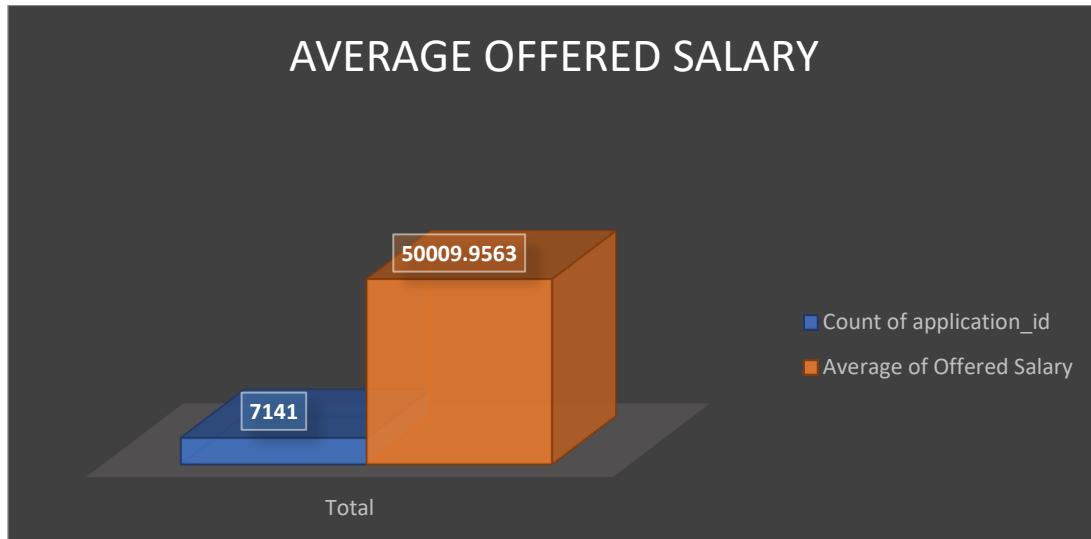


B) Salary Analysis:

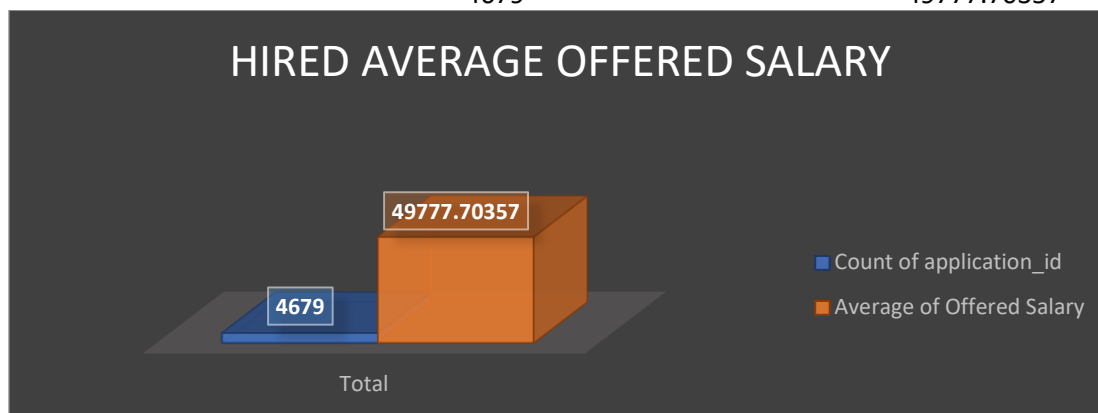
The average salary is calculated by adding up the salaries of a group of employees and then dividing the total by the number of employees.

Your Task: What is the average salary offered by this company? Use Excel functions to calculate this.

Count of application_id	Average of Offered Salary
7141	50009.9563



Count of application_id	Average of Offered Salary
4679	49777.70357



C) Salary Distribution:

Class intervals represent ranges of values, in this case, salary ranges. The class interval is the difference between the upper and lower limits of a class.

Your Task: Create class intervals for the salaries in the company. This will help you understand the salary distribution.

S13											
	A	B	C	D	E	F	G	H	I	J	K
1											
2											
3											
4											
5											
6											
7											
8											
9											
10											
11											
12											
13											
14											
15											
16											
17											
18											
19											
20											

TASK-3 Salary Distribution

5	OFFERED SALARY										
6	MAXIMUM SALARY	99967									
7	MINIMUM SALARY	100									
8											
9	HIRED SALARY										
10	MAXIMUM SALARY	99967									
11	MINIMUM SALARY	800									
12											
13	LIMIT										
14	LOWER LIMIT	0									
15	UPPER LIMIT	100000									
16											
17											
18											
19											
20											

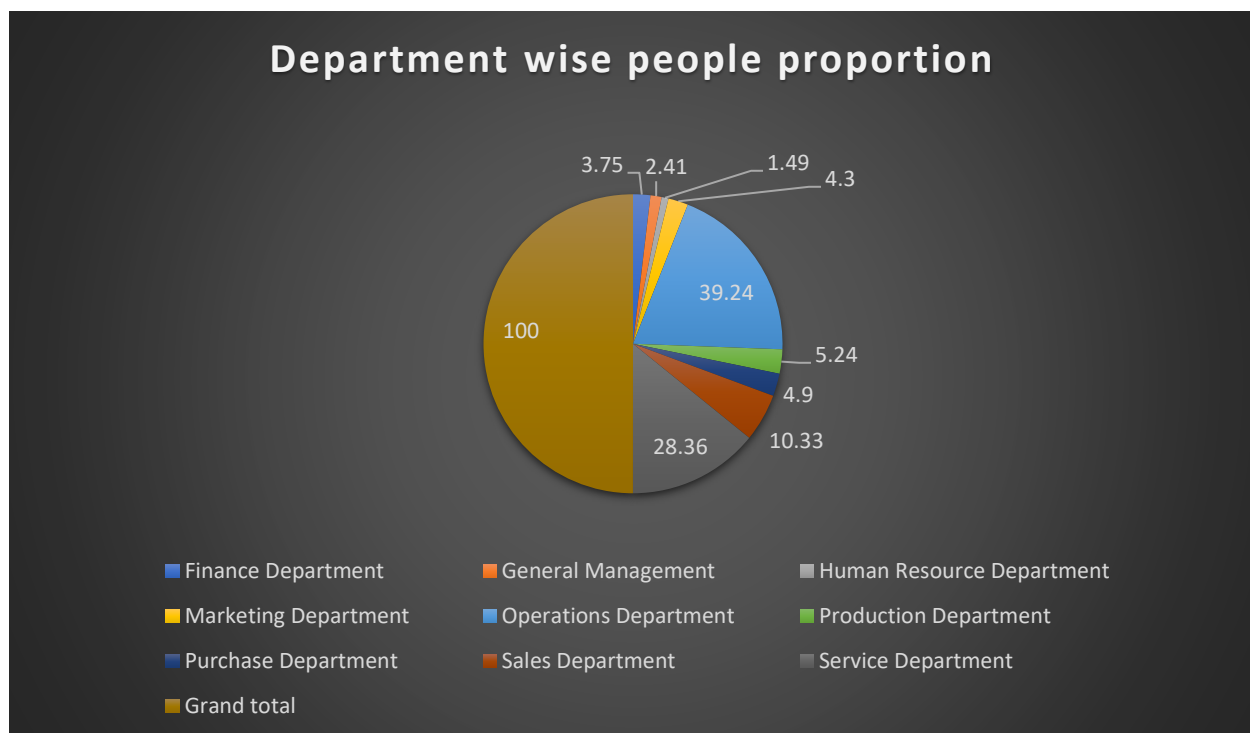


D) Departmental Analysis:

Visualizing data through charts and plots is a crucial part of data analysis.

Your Task: Use a pie chart, bar graph, or any other suitable visualization to show the proportion of people working in different departments

DEPARTMENT	PROPORTION	DEPT WISE PROPORTION
Finance Department	3.75	287
General Management	2.41	171
Human Resource Department	1.49	96
Marketing Department	4.3	325
Operations Department	39.24	2762
Production Department	5.24	379
Purchase Department	4.9	332
Sales Department	10.33	743
Service Department	28.36	2045
Grand total	100	7140

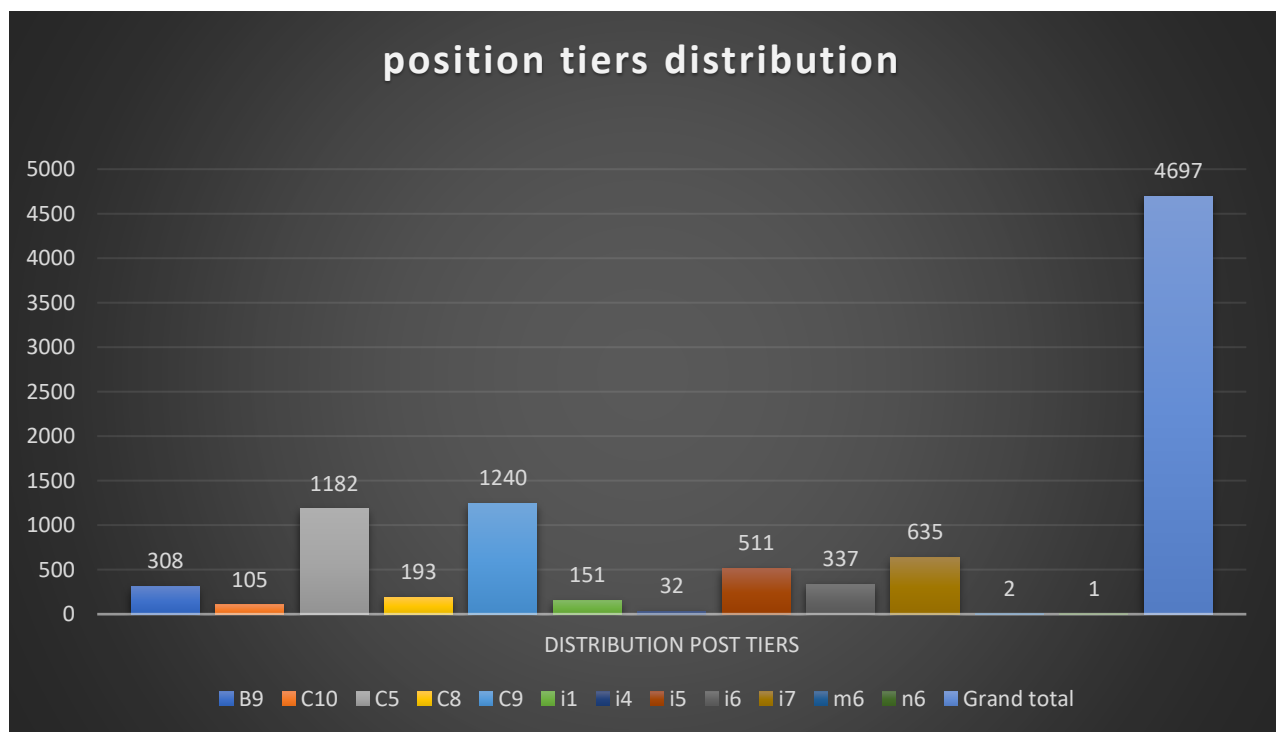


E) Position Tier Analysis:

Different positions within a company often have different tiers or levels.

Your Task: Use a chart or graph to represent the different position tiers within the company. This will help you understand the distribution of positions across different tiers.

POST NAME	DISTRIBUTION POST TIERS
B9	308
C10	105
C5	1182
C8	193
C9	1240
i1	151
i4	32
i5	511
i6	337
i7	635
m6	2
n6	1
Grand total	4697



This report summarizes the data analytics tasks performed on the dataset. The provided results offer insights into various aspects of the company, including hiring patterns, salary distribution, departmental composition, and position tiers. Further analysis and interpretation can be

conducted based on these findings to inform decision-making processes within the organization.

Result

The project yielded significant insights into the hiring process analytics, contributing to a deeper understanding of organizational dynamics. By analyzing hiring patterns, salary distribution, and departmental composition, actionable insights were obtained to optimize recruitment strategies and drive organizational growth.